

1. Decisions

Decision	Choice	Reason
Instrument & Timeframe	BTC/USDT, 15 min	- Liquid and trades 24/7 - 15m yields more potential crossovers & lower noise
MA Type	9 EMA / 21 EMA	- EMA react faster than SMA - 9/21 EMA is a standard intraday pairs on shorter timeframe - EMA is computed in-house and verified against taapi rather than depends on providers to ensure reliability
Confirmation / Debounce Rules	None	- A confirmation/debounce rule includes a filter to our strategy, which changes how the strategy works. - Filter will drop >30% of the signals, which may cause lesser/silently missed potential signals
IM Platform & Integration Method	Telegram - BotFather Bot API	- Free & easy integration and installation - Everyday use messaging app - Advanced automation capability
Hosting / Scheduling Approach	Oracle Always Free VM, Ubuntu 24.04; Anchored to wall clock & loop every 300s, +20s after each bar close	- Need an always-on server for ≥ 3 days service - Oracle offers 24/7 server for free - systemd keeps server alive and auto-restarts on any exits and reboots - Each wake is recomputed from the clock, so detection latency stays constant - Polls and loop every 300s as a safety net so that a failed poll is covered by the next - 20s delays after close to ensure exchange has published the finalised bar before fetching
State Management	JSON: last processed bar time, EMA relationship, and Exchange ID, saved after confirms delivery	- Saved only after confirmation to prevent duplication alert - A failed send will not be saved and will be retried again - Last processed bar is saved to prevent duplication alert on restarts or overlaps - Exchange id to record which exchange the prices are from and to make sure new bars are compared to the previous one from same exchange - EMA relationship is saved so that program knows what each new bar is compared against to detect a crossover - JSON is a small reliable file for a single instrument and it writes automatically so a crash won't corrupt it

2. Trade-offs

- Weaker signal observed as no confirmation/debounce rule set to receive all raw crossover signals and bars tend to reverse after 2-3 bars
- Poll every 300s (3× per bar) instead of once per bar close. Costs 3× the API calls, but buys redundancy
- REST polling instead of a WebSocket stream. Slightly higher latency, but a WebSocket's silent disconnect is a silent miss

3. What I'd do differently with more time

- Add execution layer to the bot by Integrating brokerage account api with auto trade execution feature
- Productize it. Build interactive Telegram bot with a watchlist across assets, news feed, mini trading dashboard in chat and multiple strategies
- Include a watcher outside the service to check/watch on system's status